



# On the treewidth of token and Johnson graphs

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## ABSTRACT

Let  $G$  be a graph on  $n$  vertices and  $1 \leq k \leq n$  a fixed integer. The  $k$ -token graph of  $G$  is the graph  $F_k(G)$  whose vertex set consists of all  $k$ -subsets of the vertex set of  $G$ , where two vertices  $A$  and  $B$  are adjacent in  $F_k(G)$  whenever their symmetric difference  $A \Delta B$  is an edge of  $G$ . In this paper we study the treewidth of  $F_k(G)$  when  $G$  is a star, path, or a complete graph. We show that in the first two cases, the treewidth is of order  $\Theta(n^{k-1})$ , and of order  $\Theta(n^k)$  in the third case. We conjecture that our upper bound for the treewidth of  $F_k(K_n)$  is tight. This is particularly relevant since  $F_k(K_n)$  is isomorphic to the well known Johnson graph  $J(n, k)$ .

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## 1. Introduction

Let  $G$  be a graph on  $n$  vertices, and let  $1 \leq k \leq n-1$  be an integer. The  $k$ -token graph of  $G$ , denoted by  $F_k(G)$ , is the graph whose vertex set consists of all  $k$ -subsets of vertices of  $G$ , where two vertices  $A$  and  $B$  are adjacent in  $F_k(G)$  whenever their symmetric difference  $A \Delta B$  is an edge of  $G$ . See Fig. 1 for an example. The name “token graphs” comes from the following interpretation. Consider  $k$  indistinguishable tokens placed at the vertices of  $G$ , at most one token per vertex. Create a new graph whose vertices are all possible token configurations, and make two adjacent if one token configuration can be reached from the other by sliding a token along an edge to an unoccupied vertex. The resulting graph is isomorphic to  $F_k(G)$ . This interpretation was given by Fabila-Monroy, Flores-Peñaloza, Huemer, Hurtado, Urrutia, and Wood [11]. However, token graphs have been defined independently several times and under different names (see [1,3,11,15,17,19]). We make use of the fact that  $F_k(G) \cong F_{n-k}(G)$  for every graph  $G$  on  $n$  vertices and each  $k$  with  $1 \leq k \leq n-1$ . Furthermore, it is readily seen that  $F_1(G) \cong G$ . Therefore, throughout the paper we may assume  $2 \leq k \leq n/2$ .

A central parameter in structural graph theory is the *treewidth* of a graph, which measures how close a graph is to being a tree. Many NP-hard problems become tractable when restricted to graphs of bounded treewidth [5]. Formally, a *tree decomposition* of  $G$  is a pair  $(T, \mathcal{V})$ , where  $T$  is a tree and  $\mathcal{V} = (V_t)_{t \in V(T)}$  is a family of sets  $V_t \subseteq V(G)$ , called bags, satisfying the following conditions:

1.  $V(G) = \bigcup_{t \in V(T)} V_t$ ;

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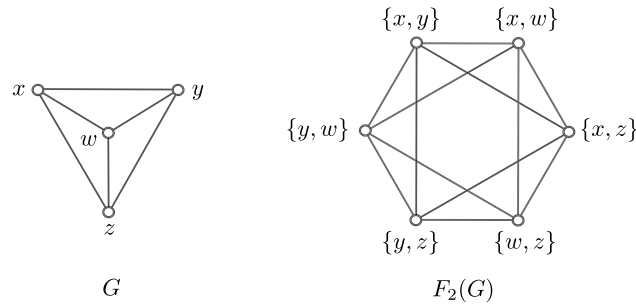


Fig. 1. A graph  $G$  and its 2-token graph  $F_2(G)$ .

2. If  $uv \in E(G)$ , then there exists  $t \in V(T)$  such that  $u, v \in V_t$ ; and
3.  $V_{t_1} \cap V_{t_3} \subseteq V_{t_2}$  whenever  $t_2$  is on the path from  $t_1$  to  $t_3$  in  $T$ .

The *width* of a tree decomposition  $(T, \mathcal{V})$  is  $\max\{|V_t| - 1 : t \in V(T)\}$ , and the *treewidth* of  $G$ , denoted  $\text{tw}(G)$ , is the minimum width over all tree decompositions of  $G$  (see Fig. 2 for an example). If the tree is restricted to be a path, the decomposition is called a *path decomposition* and the corresponding parameter is the *pathwidth* of  $G$ , denoted by  $\text{pw}(G)$ . It is clear from the definitions that  $\text{pw}(G) \geq \text{tw}(G)$ . Some basic examples include the facts that the treewidth of a tree is one, while the treewidth of the complete graph on  $n$  vertices is  $n - 1$ . Moreover, if  $H$  is a subgraph of  $G$ , then  $\text{tw}(H) \leq \text{tw}(G)$ . In this paper, we establish both lower and upper bounds for the treewidth of  $F_k(G)$  when  $G$  is a star, path, or a complete graph.

For the  $k$ -token graphs of stars and paths, we establish the following results.

**Theorem 1.** Let  $S_n$  be the star graph on  $n + 1$  vertices, and let  $2 \leq k \leq (n + 1)/2$  be a fixed integer. Then

$$\text{tw}(F_k(S_n)) = \Theta(n^{k-1}).$$

**Theorem 2.** Let  $P_n$  be the path graph on  $n$  vertices, and let  $2 \leq k \leq n/2$  be a fixed integer. Then

$$\text{tw}(F_k(P_n)) = \Theta(n^{k-1}).$$

Furthermore, we determine the exact values of  $\text{tw}(F_2(P_n))$  and  $\text{tw}(F_2(S_n))$ . Specifically, we show that

$$\text{tw}(F_2(S_n)) = n - 1 \quad \text{and} \quad \text{tw}(F_2(P_n)) = \left\lfloor \frac{n}{2} \right\rfloor.$$

We point out that the  $k$ -token graph of the complete graph  $K_n$  on  $n$  vertices is the well-known Johnson graph  $J(n, k)$ ; this graph is defined as the graph whose vertices are all the  $k$ -subsets of an  $n$ -set, and where two such subsets are adjacent whenever they intersect in exactly  $k - 1$  elements. In this paper, we establish the exact value of  $\text{tw}(F_2(K_n))$ , which is equal to its pathwidth.

**Theorem 3.** For  $n \geq 4$ ,

$$\text{pw}(F_2(K_n)) = \text{tw}(F_2(K_n)) = \begin{cases} \frac{n}{2} \left( \frac{n}{2} - 1 \right) + n - 2 & \text{when } n \text{ is even,} \\ \left( \frac{n-1}{2} \right)^2 + n - 2 & \text{when } n \text{ is odd.} \end{cases}$$

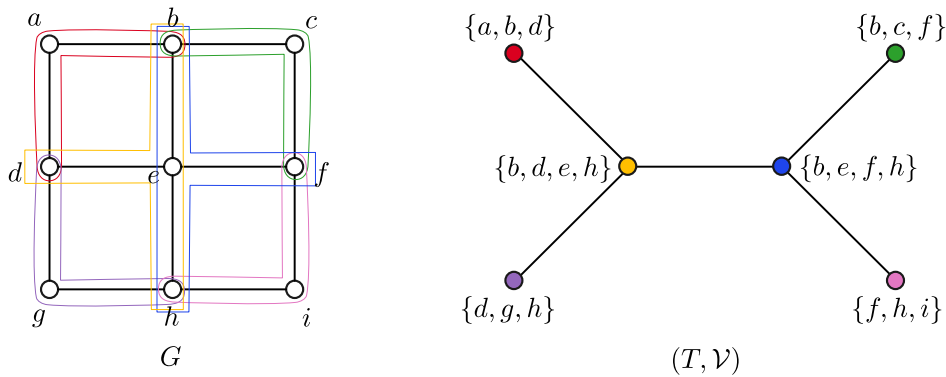
The treewidth of  $F_2(K_n)$  has been computed previously by Harvey and Wood [13]; however, we use a different method based on a result on extremal graph theory, which may be of independent interest.

We also establish lower and upper bounds for  $\text{tw}(F_k(K_n))$  when  $k \geq 3$ . For the upper bound, we provide a path decomposition of  $F_k(K_n)$  and prove the following result on its pathwidth:

**Theorem 4.** Let  $2 \leq k \leq n/2$ . Then  $\text{pw}(F_k(K_n))$  is at most

$$\max \left\{ \left\lfloor \frac{n+1}{k} \right\rfloor \binom{n - \lfloor \frac{n+1}{k} \rfloor}{k-1} + \left\lfloor \frac{n+1}{k} \right\rfloor \sum_{i=2}^k \binom{(k-i) \lfloor \frac{n+1}{k} \rfloor}{k-i} - \sum_{i=1}^{k-1} \binom{(k-i-1) \lfloor \frac{n+1}{k} \rfloor + 1}{k-i}, \right. \\ \left. \left\lceil \frac{n}{k} \right\rceil \binom{n - \lceil \frac{n}{k} \rceil}{k-1} + \left\lceil \frac{n}{k} \right\rceil \sum_{i=2}^k \binom{(k-i) \lceil \frac{n}{k} \rceil}{k-i} - \sum_{i=1}^{k-1} \binom{(k-i-1) \lceil \frac{n}{k} \rceil + 1}{k-i} \right\} - 1.$$

Further, if  $k$  divides  $n$  or  $n + 1$ , these two values are equal.



**Fig. 2.** A graph  $G$  (left) and an optimal tree decomposition  $(T, \mathcal{V})$  (right). The colors in the decomposition indicate the bags that contain the respective vertices in  $G$ , ensuring that  $\text{tw}(G) = 3$ .

For the case  $k = 3$ , we provide the following upper bound, which is consistent with [Theorem 4](#).

**Corollary 5.** *Let  $n \geq 6$ . Then*

$$\text{pw}(F_3(K_n)) \leq \left\lceil \frac{n}{3} \right\rceil \binom{n - \left\lceil \frac{n}{3} \right\rceil}{2} + \frac{1}{2} \left\lceil \frac{n}{3} \right\rceil \left( \left\lceil \frac{n}{3} \right\rceil + 1 \right) - 2.$$

It is worth mentioning that through computational experimentation, we have confirmed that the upper bound given in [Lemma 19](#) is tight for certain small values of  $n$  with respect to the treewidth. Although the result provides an upper bound on the pathwidth, this confirms that the treewidth matches the bound in these cases. These observations lead us to formulate [Conjecture 21](#) in [Section 5](#), which asserts that the upper bound established in [Theorem 4](#) is optimal for all  $n$  and  $k$  with  $2 \leq k \leq n/2$ .

This paper is organized as follows. In [Section 2](#), we present the proofs of [Theorems 1](#) and [2](#), concerning the treewidth of token graphs of stars and paths. [Section 3](#) contains the proof of [Theorem 3](#), which establishes the exact treewidth and pathwidth of  $F_2(K_n)$ . In [Section 4](#), we address [Theorem 4](#) and [Lemma 19](#), concerning upper bounds for the treewidth of Johnson graphs. Finally, in [Section 5](#), we present concluding remarks and propose two conjectures.

To prove our results, we make use of fundamental tools from structural and spectral graph theory. These include brambles and their connection to treewidth, the algebraic connectivity of graphs, and spectral properties of token graphs. We briefly recall the necessary background and notation. For standard terminology, we refer the reader to [\[6\]](#). For a finite set  $X$ , we denote by  $\binom{X}{k}$  the family of all  $k$ -subsets of  $X$ . Throughout the paper, we assume  $V(G) = [n] := \{1, \dots, n\}$  unless stated otherwise, so  $V(F_k(G)) = \binom{[n]}{k}$ .

We say that two subsets of  $V(G)$  *touch* if either they have a vertex in common, or there is an edge between them in  $G$ . A *bramble*,  $\mathcal{B}$ , is a family of pairwise touching subsets of vertices of  $G$ , each of which induces a connected subgraph. A subset  $S$  of vertices of  $G$  is said to *cover*  $\mathcal{B}$ , if every set of  $\mathcal{B}$  contains a vertex in  $S$ ; we call  $S$  a *hitting set* for  $\mathcal{B}$ . The order of  $\mathcal{B}$ , denoted  $\|\mathcal{B}\|$ , is the least number of vertices in a hitting set of  $\mathcal{B}$ . The *bramble number* of  $G$ , denoted by  $\text{bn}(G)$ , is the maximum order of a bramble of  $G$ . Seymour and Thomas found the following duality theorem, which relates brambles and treewidth.

**Theorem 6** ([\[18\]](#)). *For any graph  $G$ ,*

$$\text{tw}(G) = \text{bn}(G) - 1.$$

We also use a spectral approach based on the algebraic connectivity of a graph. Let  $L(G) = D(G) - A(G)$  be the Laplacian matrix of  $G$ , where  $D(G)$  is the degree matrix and  $A(G)$  the adjacency matrix. The second smallest eigenvalue of  $L(G)$ , denoted  $\lambda_2(G)$ , is called the *algebraic connectivity*. Chandran and Subramanian showed the following relationship between algebraic connectivity and treewidth (see also [\[2\]](#)).

**Theorem 7** ([\[8\]](#)). *If  $G$  is a connected graph with maximum degree  $\Delta$ , then*

$$\frac{|V|}{12\Delta} \lambda_2(G) - 1 \leq \text{tw}(G).$$

The Laplacian spectrum of token graphs was studied by Dalfó, Duque, Fabila-Monroy, Fiol, Huemer, Trujillo-Negrete, and Zaragoza-Martínez [\[9\]](#). The authors conjectured that the equality

$$\lambda_2(F_k(G)) = \lambda_2(G) \tag{1}$$

always holds; and proved it for certain families of graphs. However, unbeknownst to the authors this is a special case of Aldous' spectral gap conjecture. This conjecture was proved by Caputo, Liggett and Richthammer [7]. We use (1) and Theorem 7 to obtain lower bounds for the treewidth of the token graph of paths and complete graphs.

## 2. Stars and paths

The star,  $S_n$ , of  $n$  leaves is the complete bipartite graph  $K_{1,n}$ . Throughout this section, we assume that  $V(S_n) = \{0\} \cup [n]$ , where vertex 0 is the center of  $S_n$ . We also use  $P_n$  to denote the path  $(1, \dots, n)$ .

Our first result is for the treewidth of token graphs of stars, namely Theorem 1:

**Theorem 1.** *Let  $S_n$  be the star graph on  $n + 1$  vertices, and let  $2 \leq k \leq (n + 1)/2$  be a fixed integer. Then*

$$\text{tw}(F_k(S_n)) = \Theta(n^{k-1}).$$

**Proof.** As mentioned in the Introduction, we have  $F_k(S_n) \cong F_{n+1-k}(S_n)$ , so without loss of generality, we may assume  $k \leq \frac{n+1}{2}$ . In what follows, we show the following bounds:

$$\frac{1}{12} \cdot \frac{n^{k-1}}{k!} + \Theta(n^{k-2}) \leq \text{tw}(F_k(S_n)) \leq \binom{n}{k-1} - 1 = k \cdot \frac{n^{k-1}}{k!} + \Theta(n^{k-2}).$$

This directly implies that  $\text{tw}(F_k(S_n)) = \Theta(n^{k-1})$ , as desired.

The graph  $F_k(S_n)$  is semiregular (see Lemma 3.8(i) in [4]): its vertices have degree either  $k$  or  $n - (k - 1)$ , depending on whether they contain the central vertex 0. Since we are assuming  $k \leq \frac{n+1}{2}$ , the maximum degree of  $F_k(S_n)$  is  $\Delta(F_k(S_n)) = n - (k - 1)$ . Additionally, it is well known that  $\lambda_2(S_n) = 1$  (see [10]). Therefore, by (1) and Theorem 7, we obtain

$$\begin{aligned} \text{tw}(F_k(S_n)) &\geq \frac{1}{12(n - (k - 1))} \binom{n+1}{k} - 1 \\ &= \frac{1}{12} \cdot \frac{n^{k-1}}{k!} + \Theta(n^{k-2}). \end{aligned}$$

To prove the upper bound we construct a tree decomposition  $(T, \mathcal{V})$  of  $F_k(S_n)$ . Let  $T$  be a star graph with 0 as its center vertex, and such that every  $A \in \binom{[n]}{k}$  is a leaf of  $T$ . That is,  $V(T) = \{0\} \cup \binom{[n]}{k}$ , and each vertex  $A \in \binom{[n]}{k}$  is adjacent to the center 0. Let

$$V_0 := \{B \in V(F_k(S_n)) : 0 \in B\} \quad \text{and} \quad V_A := \{A\} \cup \{(A \setminus \{a_i\}) \cup \{0\} : a_i \in A\},$$

for every  $A \in \binom{[n]}{k}$ . Finally, let  $\mathcal{V} := (V_t)_{t \in V(T)}$ . See Fig. 3 for an example of this construction. We have the following.

1. Every vertex of  $F_k(S_n)$  is contained in some  $V_A$ .
2. Each edge in  $F_k(S_n)$  corresponds to a pair of sets of the form  $A$  and  $(A \setminus \{a_i\}) \cup \{0\}$ , where  $A \in \binom{[n]}{k}$  and  $a_i \in A$ . Thus,

$$A, (A \setminus \{a_i\}) \cup \{0\} \in V_A.$$

3. For every pair  $A, A' \in \binom{[n]}{k}$ , if  $|A \cap A'| \leq k - 2$ , then  $V_A \cap V_{A'} = \emptyset \subset V_0$ , and if  $|A \cap A'| = k - 1$ , then

$$V_A \cap V_{A'} = \{(A \cap A') \cup \{0\}\} \subset V_0.$$

Therefore,  $(T, \mathcal{V})$  is a tree decomposition of  $F_k(S_n)$ . Its width is equal to

$$\binom{n}{k-1} - 1 = \frac{1}{(k-1)!} n^{k-1} + \Theta(n^{k-2}). \quad \square$$

We now compute the exact value of the treewidth of the 2-token graph of a star.

### Proposition 8.

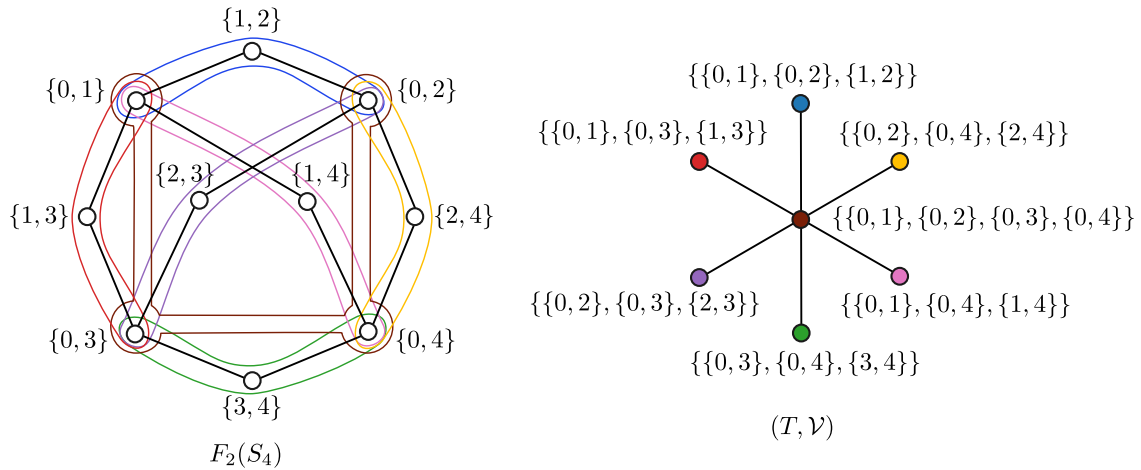
$$\text{tw}(F_2(S_n)) = n - 1.$$

**Proof.** As shown in the proof of Theorem 1, we have

$$\text{tw}(F_2(S_n)) \leq \binom{n}{1} - 1 = n - 1.$$

For the lower bound, we construct a bramble of  $F_2(S_n)$  in which each hitting set has at least  $n$  vertices. For every  $1 \leq i \leq n$ , let

$$B_i := \{\{s, i\} : 0 \leq s < i\}.$$



**Fig. 3.** The 2-token graph  $F_2(S_4)$  of the star  $S_4$  (left) and an optimal tree decomposition  $(T, \mathcal{V})$  (right). The colors in the decomposition indicate the bags that contain the respective vertices in  $F_2(S_4)$ .

Each  $B_i$  induces a connected subgraph of  $F_2(S_n)$ . Moreover, for every  $1 \leq i < j \leq n$ , the pair  $\{\{0, i\}, \{i, j\}\}$  is an edge of  $F_2(S_n)$  with one endpoint in  $B_i$  and the other in  $B_j$ , so the sets  $B_i$  and  $B_j$  are touching. Thus, the collection

$$\mathcal{B} := \{B_i : 1 \leq i \leq n\},$$

is a bramble of  $F_2(S_n)$ . Since  $B_i \cap B_j = \emptyset$ , any hitting set of  $\mathcal{B}$  must contain at least one vertex from each  $B_i$ , and therefore must have cardinality at least  $n$ . By Theorem 6, we conclude that

$$\text{tw}(F_2(S_n)) \geq n - 1. \quad \square$$

We now turn our attention to token graphs of paths. To analyze them, we use the notion of Cartesian products. Let  $G_1, \dots, G_n$  be graphs. The *Cartesian product* of  $G_1, \dots, G_n$  is the graph  $G_1 \square \dots \square G_n$  with vertex set  $V(G_1) \times \dots \times V(G_n)$ , where  $(x_1, \dots, x_n)$  is adjacent to  $(y_1, \dots, y_n)$  if and only if there exists an index  $1 \leq i \leq n$  such that  $x_i$  is adjacent to  $y_i$  and  $x_j = y_j$  for all  $j \neq i$ .

**Lemma 9.** If  $G_1, \dots, G_k$  are vertex-disjoint subgraphs of  $G$ , then

$$\text{tw}(F_k(G)) \geq \text{tw}(G_1 \square \dots \square G_k).$$

**Proof.** Consider the subgraph of  $F_k(G)$  induced by the token configurations where exactly one token is placed in each  $G_i$ . This subgraph is isomorphic to  $G_1 \square \dots \square G_k$ , which implies the desired inequality.  $\square$

To obtain a lower bound for the treewidth of  $F_k(P_n)$ , we use the following result of Hickingbotham and Wood [14] regarding the treewidth of  $d$ -dimensional grids.

**Theorem 10** ([14]). Let  $d \geq 2$  be a fixed integer, and let  $n_1 \geq \dots \geq n_d \geq 1$ . Then

$$\text{tw}(P_{n_1} \square \dots \square P_{n_d}) = \Theta \left( \prod_{j=2}^d n_j \right).$$

We are now ready to prove Theorem 2, which we restate for completeness.

**Theorem 2.** Let  $P_n$  be the path graph on  $n$  vertices, and let  $2 \leq k \leq n/2$  be a fixed integer. Then

$$\text{tw}(F_k(P_n)) = \Theta(n^{k-1}).$$

**Proof.** We begin by showing that  $\text{tw}(F_k(P_n)) = \Omega(n^{k-1})$ . Consider a partition of  $P_n$  into  $k$  vertex-disjoint subpaths  $P'_1, \dots, P'_k$ , each with  $\lfloor n/k \rfloor$  consecutive vertices. Specifically, for each  $i = 1, \dots, k$ , define

$$V(P'_i) := \left\{ (i-1) \left\lfloor \frac{n}{k} \right\rfloor + 1, \dots, i \left\lfloor \frac{n}{k} \right\rfloor \right\}.$$

Since  $k$  is fixed, [Lemma 9](#) and [Theorem 10](#) imply

$$\text{tw}(F_k(P_n)) = \Omega\left(\left\lfloor \frac{n}{k} \right\rfloor^{k-1}\right) = \Omega(n^{k-1}).$$

Next, to establish the upper bound, we argue that  $\text{tw}(F_k(P_n)) = O(n^{k-1})$ . Define the graph  $H$  as the Cartesian product of  $k$  copies of  $P_{n-(k-1)}$ , with  $V(P_{n-(k-1)}) := \{1, \dots, n - k + 1\}$ . Define  $H'$  as the subgraph of  $H$  induced by all  $k$ -tuples  $(y_1, \dots, y_k)$  with  $y_1 \leq y_2 \leq \dots \leq y_k$ . Now, define the map  $f : V(F_k(P_n)) \rightarrow V(H')$  as follows. For each vertex  $A = \{x_1, \dots, x_k\} \in V(F_k(P_n))$ , with  $x_1 < \dots < x_k$ , define

$$f(A) := (y_1, \dots, y_k), \text{ where } y_i = x_i - (i - 1) \text{ for each } i = 1, \dots, k.$$

We will show that  $f$  is an isomorphism. It is readily seen that  $f$  is a bijection from  $V(F_k(P_n))$  to  $V(H')$ , so it remains to show that  $f$  preserves adjacencies and non-adjacencies. Fix a vertex  $A = \{x_1, \dots, x_k\}$  in  $F_k(P_n)$ , and let  $B$  be another vertex of  $F_k(P_n)$ . Observe that  $A$  and  $B$  are adjacent in  $F_k(P_n)$  if and only if there exists  $i \in [k]$  such that  $B = (A \setminus \{x_i\}) \cup \{x'_i\}$ , where  $x'_i = x_i - 1$  or  $x'_i = x_i + 1$ . This adjacency condition holds if and only if  $f(A)(j) = f(B)(j)$  for all  $j \neq i$ , whereas  $f(A)(i) = x_i - (i - 1)$  and  $f(B)(i) = x'_i - (i - 1)$ , meaning that  $f(A)(i)$  and  $f(B)(i)$  are adjacent in the  $i$ th copy of  $P_{n-(k-1)}$ . This occurs precisely when  $f(A)$  and  $f(B)$  are adjacent in  $H'$ . Thus,  $AB$  is an edge of  $F_k(P_n)$  if and only if  $f(A)f(B)$  is an edge of  $H'$ . Therefore,  $f$  is an isomorphism from  $F_k(P_n)$  to  $H'$ .

Finally, since  $k$  is fixed and  $F_k(P_n)$  is isomorphic to the subgraph  $H'$  of  $H$ , [Theorem 10](#) implies that

$$\text{tw}(F_k(P_n)) = \text{tw}(H') \leq \text{tw}(H) = O((n - (k - 1))^{k-1}) = O(n^{k-1}).$$

The result follows.  $\square$

We now focus on establishing the exact treewidth of  $F_2(P_n)$ . To this end, we use a combinatorial characterization of treewidth due to Lucena [16], which is based on vertex orderings and *borders*. Given a graph  $G$  and a subset  $S \subseteq V(G)$ , the subgraph of  $G$  induced by  $S$  is denoted by  $G[S]$ . The *border* of  $S$ , denoted by  $\beta(S)$ , is defined as follows.

- If  $G[S]$  is connected, then  $\beta(S) = |N(S) \setminus S|$ , where  $N(S)$  denotes the neighborhood of  $S$ ;
- If  $G[S]$  is disconnected, then  $\beta(S) = \max_{C \in \mathcal{C}} \beta(C)$ , where  $\mathcal{C}$  is the set of components of  $G[S]$ .

An ordering  $\pi$  of the vertices of  $G$  is a bijection from  $[n]$  to  $V(G)$ . Let  $\pi$  be a fixed ordering of the vertices of  $V(G)$ . For  $i \in [n]$ , let  $T_i := \{\pi(1), \dots, \pi(i)\}$ . The *max border* of  $G$  with respect to  $\pi$ , denoted by  $MB_\pi(G)$ , is defined as

$$MB_\pi(G) := \max_{i \in [n]} \beta(T_i).$$

Let the *minimax border size* of a graph  $G$  be given by

$$MMB(G) := \min_{\text{orderings } \pi} MB_\pi(G).$$

Lucena proved the following characterization:

**Theorem 11** ([16]). *The minimax border size of  $G$  equals its treewidth.*

We now use this result to compute the exact value of the treewidth of the 2-token graph of a path.

**Proposition 12.**

$$\text{tw}(F_2(P_n)) = \left\lfloor \frac{n}{2} \right\rfloor.$$

**Proof.** We begin by establishing the lower bound. Consider the subgraph  $H$  of  $F_2(P_n)$  induced by the vertex set

$$\left\{ \{x_1, x_2\} \in V(F_2(P_n)) : 1 \leq x_1 \leq \left\lfloor \frac{n}{2} \right\rfloor < x_2 \leq n \right\}.$$

It is straightforward to observe that  $H$  is isomorphic to  $P_{\lfloor \frac{n}{2} \rfloor} \square P_{\lceil \frac{n}{2} \rceil}$ . By applying [Lemma 9](#) and the fact that  $\text{tw}(P_q \square P_r) = \min\{q, r\}$  for all integers  $q, r \geq 1$ , we obtain the lower bound:

$$\text{tw}(F_2(P_n)) \geq \left\lfloor \frac{n}{2} \right\rfloor.$$

To show the upper bound, we define an ordering  $\pi$  on  $V(F_2(P_n))$  and apply [Theorem 11](#). For  $3 \leq i \leq 2n - 1$ , let

$$\mathcal{D}_i := \{\{x_1, x_2\} \in F_2(P_n) : x_1 + x_2 = i\};$$

see [Fig. 4](#) for an example. Note that  $|\mathcal{D}_i|$  is maximized when  $i = n + 1$ , and in such a case, we have  $|\mathcal{D}_i| = \lfloor \frac{n}{2} \rfloor$ . For every pair of vertices  $\{x_1, x_2\}, \{y_1, y_2\} \in V(F_2(P_n))$  with  $x_1 < x_2$  and  $y_1 < y_2$ , we say that

$$\{x_1, x_2\} \leq \{y_1, y_2\} \text{ whenever } x_1 + x_2 < y_1 + y_2, \text{ or } x_1 + x_2 = y_1 + y_2 \text{ and } x_1 < y_1.$$

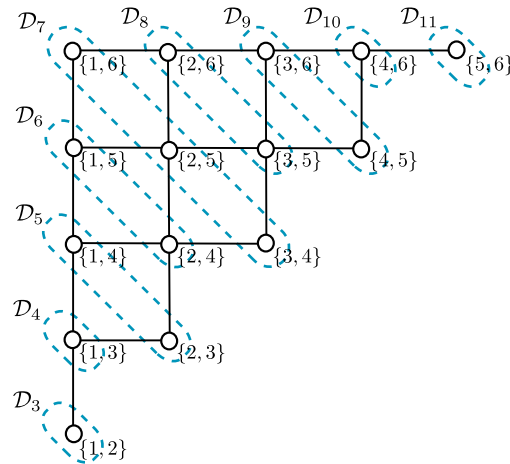


Fig. 4. Subsets  $\mathcal{D}_i$  in  $F_2(P_6)$ .

Let  $\pi$  be the corresponding ordering of  $V(F_2(P_n))$ . Let  $1 \leq r \leq \binom{n}{2}$  be an integer. Recall that  $T_r$  is the set of the first  $r$  vertices in the order  $\pi$ , i.e.,  $T_r = \{\pi(1), \dots, \pi(r)\}$ . Note that the subgraph of  $F_2(P_n)$  induced by  $T_r$  is connected, and thus,  $\beta(T_r) = |N(T_r) \setminus T_r|$ .

Let  $i$  be the maximum index such that  $\mathcal{D}_1 \cup \dots \cup \mathcal{D}_i \subseteq T_r$ .

- If  $\pi(r) \in \mathcal{D}_i$ , then  $\beta(T_r) = |N(T_r) \setminus T_r| = |\mathcal{D}_{i+1}|$  and  $\beta(T_r) \leq \lfloor \frac{n}{2} \rfloor$ .
- Suppose that  $\pi(r) \in \mathcal{D}_{i+1}$ ; thus,  $\pi(r) = \{j, i+1-j\}$  for some  $j$ . If  $i < n+1$ , then  $|(N(T_r) \setminus T_r) \cap \mathcal{D}_{i+2}| = j+1$  and  $|(N(T_r) \setminus T_r) \cap \mathcal{D}_{i+1}| = |\mathcal{D}_{i+1}| - j < \lfloor \frac{n}{2} \rfloor - j$ . Thus,  $\beta(T_r) \leq \lfloor \frac{n}{2} \rfloor$ . Suppose that  $i \geq n+1$ . We have that  $|(N(T_r) \setminus T_r) \cap \mathcal{D}_{i+2}| = j$  and  $|(N(T_r) \setminus T_r) \cap \mathcal{D}_{i+1}| = |\mathcal{D}_{i+1}| - j \leq \lfloor \frac{n}{2} \rfloor - j$ . Thus,  $\beta(T_r) \leq \lfloor \frac{n}{2} \rfloor$ .

Therefore,

$$\beta(T_r) \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

By Theorem 11 we have that

$$\text{tw}(F_2(P_n)) \leq MB_\pi(F_2(P_n)) = \max_{i \in [n]} \beta(T_i) = \left\lfloor \frac{n}{2} \right\rfloor. \quad \square$$

### 3. Treewidth of $F_2(K_n)$

In this section, we aim to establish Theorem 3. We first provide an upper bound by constructing a path decomposition  $(P, \mathcal{V})$  of  $F_2(K_n)$  whose width matches the value stated in Theorem 3. For the lower bound, we exhibit a bramble  $\mathcal{B}$  of  $F_2(K_n)$  whose order is exactly the value given in Theorem 3 plus one. For the reader's convenience, we restate Theorem 3.

**Theorem 3.** For  $n \geq 4$ ,

$$\text{pw}(F_2(K_n)) = \text{tw}(F_2(K_n)) = \begin{cases} \frac{n}{2} \left( \frac{n}{2} - 1 \right) + n - 2 & \text{when } n \text{ is even,} \\ \left( \frac{n-1}{2} \right)^2 + n - 2 & \text{when } n \text{ is odd.} \end{cases}$$

As mentioned in the Introduction, the treewidth of  $F_2(K_n)$  was previously computed by Harvey and Wood [13] for the line graph of complete graphs, which is isomorphic to  $F_2(K_n)$  and also corresponds to the Johnson graph  $J(n, 2)$ . For the upper bound in Theorem 3, we present the same path decomposition as the one used by Harvey and Wood [13], which we include here for completeness and because it will be generalized in the next section for the case  $k \geq 3$ . For the lower bound, however, we construct a different bramble, whose order is determined by a result of Faudree and Schelp on extremal graph theory.

Throughout this section we assume that whenever  $\{i, j\}$  is a vertex of  $F_2(K_n)$ , we have that  $i < j$ . We construct the path decomposition  $(P, \mathcal{V})$  of  $F_2(K_n)$  as follows. Let  $P$  be the path with vertex set  $[n]$ , where each vertex  $l$  is adjacent to  $l+1$  for  $1 \leq l < n$ . For every  $1 \leq l \leq n$ , let

$$V_l := \{\{i, j\} \in V(F_2(K_n)) : i \leq l \leq j\}.$$

Let

$$\mathcal{V} := (V_l)_{l \in V(P)}.$$

We have the following.

1. For every  $\{i, j\} \in V(F_2(K_n))$ , we have that  $\{i, j\} \in V_j$ . Thus,

$$V(F_2(K_n)) = \bigcup_{l \in V(P)} V_l.$$

2. If  $\{i, j\}\{i', j'\}$  is an edge of  $F_2(K_n)$ , and  $l$  is the only element in  $\{i, j\} \cap \{i', j'\}$ , then

$$\{i, j\}, \{i', j'\} \in V_l.$$

3. For every  $1 \leq l_1 < l_3 \leq n$  and  $l_1 \leq l_2 \leq l_3$ , if  $\{i, j\}$  is a vertex in  $V_{l_1} \cap V_{l_3}$ , then  $i \leq l_1 < l_3 \leq j$ , which implies  $i \leq l_2 \leq j$ , and so  $\{i, j\} \in V_{l_2}$ . Thus,

$$V_{l_1} \cap V_{l_3} \subset V_{l_2}.$$

Therefore,  $(P, \mathcal{V})$  is a path decomposition of  $F_2(K_n)$ .

For every  $1 \leq l \leq n$ , we have that

$$|V_l| = (l-1)(n-l) + n - 1.$$

This is maximized for  $l = \lfloor (n+1)/2 \rfloor$ . Therefore,

$$\text{tw}(F_2(K_n)) \leq \text{pw}(F_2(K_n)) \leq \begin{cases} \frac{n}{2} \left( \frac{n}{2} - 1 \right) + n - 2 & \text{when } n \text{ is even,} \\ \left( \frac{n-1}{2} \right)^2 + n - 2 & \text{when } n \text{ is odd.} \end{cases} \quad (2)$$

For the lower bound on  $\text{tw}(F_2(K_n))$  we make use of brambles and the following result on extremal graph theory due to Faudree and Schelp [12].

**Theorem 13** ([12]). *If  $G$  is a graph with  $|V(G)| = kt + r$ ,  $0 \leq r < k$ , containing no path with  $k + 1$  vertices, then  $|E(G)| \leq t \binom{k}{2} + \binom{r}{2}$ .*

**Proposition 14.** *For  $n \geq 4$  we have that*

$$\text{bn}(F_2(K_n)) \geq \begin{cases} \frac{n}{2} \left( \frac{n}{2} - 1 \right) + n - 1 & \text{when } n \text{ is even,} \\ \left( \frac{n-1}{2} \right)^2 + n - 1 & \text{when } n \text{ is odd.} \end{cases}$$

**Proof.**

- Suppose that  $n$  is odd. Note that the vertices of  $F_2(K_n)$  are precisely the edges of  $K_n$ . Let

$$\mathcal{B} := \left\{ B \subset V(F_2(K_n)) : B \text{ induces a path of } K_n \text{ with } \frac{n+1}{2} \text{ vertices} \right\}.$$

Note that every element of  $\mathcal{B}$  induces a connected subgraph of  $F_2(K_n)$ . Let  $B_1, B_2 \in \mathcal{B}$ . Since,  $\left| \bigcup_{A \in B_1} A \right| = \left| \bigcup_{A \in B_2} A \right| = \frac{n+1}{2}$ , we have that  $\left( \bigcup_{A \in B_1} A \right) \cap \left( \bigcup_{A \in B_2} A \right) \neq \emptyset$ . Let  $a \in \left( \bigcup_{A \in B_1} A \right) \cap \left( \bigcup_{A \in B_2} A \right)$ , and let  $b_1, b_2 \in V(K_n)$  such that  $\{a, b_1\} \in B_1$  and  $\{a, b_2\} \in B_2$ . If  $b_1 = b_2$ , then  $B_1$  and  $B_2$  have a common vertex. If  $b_1 \neq b_2$ , then the edge  $\{a, b_1\}\{a, b_2\}$  belongs to  $F_2(K_n)$  and connects a vertex in  $B_1$  to a vertex in  $B_2$ . Therefore,  $B_1$  touches  $B_2$ , and  $\mathcal{B}$  is a bramble.

Let  $S$  be a hitting set of  $\mathcal{B}$  of minimum cardinality, and let  $S^c$  be its complement in  $V(F_2(K_n))$ . Let  $H$  be the subgraph of  $K_n$  induced by  $S^c$ . If  $H$  contains a path  $P$  with  $\frac{n+1}{2}$  vertices, then we have a contradiction, since  $E(P) \in \mathcal{B}$  and  $E(P) \cap S = \emptyset$ . Thus,  $H$  contains no path with  $\frac{n+1}{2}$  vertices. By Theorem 13, with  $G = H$ ,  $k = \frac{n-1}{2}$ ,  $t = 2$  and  $r = 1$ , we have that

$$|S^c| = |E(H)| \leq 2 \binom{\frac{n-1}{2}}{2}.$$

This implies that

$$\text{bn}(F_2(K_n)) \geq |S| = \binom{n}{2} - |S^c| \geq \binom{n}{2} - 2 \binom{\frac{n-1}{2}}{2} = \left( \frac{n-1}{2} \right)^2 + n - 1,$$

as required.



- Suppose that  $n$  is even. Let

$$V_1 := \{A \in V(F_2(K_n)) : n \notin A\} \text{ and } V_2 := \{A \in V(F_2(K_n)) : n \in A\}.$$

Let  $H_1$  and  $H_2$  be the subgraphs of  $F_2(K_n)$  induced by  $V_1$  and  $V_2$ , respectively. Note that  $H_1 \cong F_2(K_{n-1})$  and  $H_2 \cong F_1(K_{n-1}) \cong K_{n-1}$ .

Let  $\mathcal{B}_1$  be a bramble of  $H_1$  defined as in the previous case, and define

$$\mathcal{B}_2 := \left\{ B \subset V_2 : |B| = \frac{n}{2} \right\}.$$

Note that  $\mathcal{B}_2$  is a bramble in  $H_2$ , since  $H_2$  is a complete graph. Define  $\mathcal{B} := \mathcal{B}_1 \cup \mathcal{B}_2$ . Let  $B_1, B_2 \in \mathcal{B}$ . Since both  $\mathcal{B}_1$  and  $\mathcal{B}_2$  are brambles, the subgraphs induced by  $B_1$  and  $B_2$  are connected. If  $B_1$  and  $B_2$  are both in  $\mathcal{B}_1$ , or are both in  $\mathcal{B}_2$ , then they touch each other. Now suppose that  $B_1 \in \mathcal{B}_1$  and  $B_2 \in \mathcal{B}_2$ . Note that  $|\bigcup_{A \in \mathcal{B}_1} A| = \frac{n}{2}$  and  $|\bigcup_{A \in \mathcal{B}_2} A| = \frac{n}{2} + 1$ .

Therefore,  $(\bigcup_{A \in \mathcal{B}_1} A) \cap (\bigcup_{A \in \mathcal{B}_2} A) \neq \emptyset$ . Let  $a \in (\bigcup_{A \in \mathcal{B}_1} A) \cap (\bigcup_{A \in \mathcal{B}_2} A)$ , and let  $b_1 \in V(K_n)$  such that  $\{a, b_1\} \in \mathcal{B}_1$ . Since all vertices in  $\mathcal{B}_2$  contain  $n$ , we have that  $\{a, n\} \in \mathcal{B}_2$ . Then, the edge  $\{a, b_1\}\{a, n\}$  belongs to  $F_2(K_n)$  and connects a vertex of  $B_1$  to a vertex of  $B_2$ . Thus,  $B_1$  and  $B_2$  touch. Therefore,  $\mathcal{B}$  is a bramble for  $F_2(K_n)$ .

Let  $S$  be a hitting set of  $\mathcal{B}$  of minimum cardinality; let  $S_1 := S \cap V_1$  and  $S_2 := S \cap V_2$ . Note that  $S_1$  is a hitting set of  $\mathcal{B}_1$  and  $S_2$  is a hitting set of  $\mathcal{B}_2$ . Therefore,

$$|S_1| \geq \left(\frac{n-2}{2}\right)^2 + n - 2. \quad (3)$$

Suppose that  $|S_2| < \frac{n}{2}$ . Since every element of  $S_2$  contains  $n$ , we have that

$$\left| \left( \bigcup_{A \in S_2} A \right) \setminus \{n\} \right| < \frac{n}{2}.$$

Thus, there exists a set  $B' \subset V(K_n) \setminus \left( \left( \bigcup_{A \in S_2} A \right) \setminus \{n\} \right)$  such that  $|B'| = \frac{n}{2}$ . Let  $B := \{\{a, n\} : a \in B'\}$ . We have a contradiction since  $B \in \mathcal{B}_2$  and  $S_2 \cap B = \emptyset$ . Therefore,

$$|S_2| \geq \frac{n}{2}. \quad (4)$$

Combining (3) and (4), we have that

$$\text{bn}(F_2(K_n)) \geq |S| = |S_1| + |S_2| \geq \left(\frac{n-2}{2}\right)^2 + n - 2 + \frac{n}{2} = \frac{n}{2} \left(\frac{n}{2} - 1\right) + n - 1,$$

as required.  $\square$

Since  $\text{tw}(F_2(K_n)) = \text{bn}(F_2(K_n)) - 1$  by Theorem 6, we derive Theorem 3 as a consequence of (2) and Proposition 14.

#### 4. Treewidth of $F_k(K_n)$

In this section, we establish an asymptotic lower bound for  $\text{tw}(F_k(K_n))$ . Additionally, we generalize the upper bound for  $\text{pw}(F_k(K_n))$  presented in Section 3, extending it from the case  $k = 2$  to arbitrary values of  $k$ .

**Proposition 15.** *If  $2 \leq k \leq n/2$  is a fixed integer, then*

$$\text{tw}(F_k(K_n)) \geq \frac{1}{12k(k!)} n^k + \Theta(n^{k-1})$$

**Proof.** The graph  $F_k(K_n)$  is  $k(n-k)$ -regular (see [4]), so  $\Delta(F_k(K_n)) = k(n-k)$ . It is well known that  $\lambda_2(K_n) = n$  (see [10]). Then, from equality (1) and Theorem 7, we have

$$\begin{aligned} \text{tw}(F_k(K_n)) &\geq \frac{n}{12k(n-k)} \binom{n}{k} - 1 \\ &= \left( \frac{n}{12k(n-k)} \right) \left( \frac{n^k}{k!} + \Theta(n^{k-1}) \right) \\ &= \frac{n^k}{12k(k!)} \left( 1 + \frac{k}{n-k} \right) + \Theta(n^{k-1}) \\ &= \frac{1}{12k(k!)} \cdot n^k + \Theta(n^{k-1}). \quad \square \end{aligned}$$

As a direct consequence of [Proposition 15](#), and considering that  $\text{tw}(F_k(K_n))$  is trivially bounded above by the number of vertices of  $F_k(K_n)$  minus 1, which is  $\binom{n}{k} - 1$ , we obtain that for a fixed integer  $k$ ,  $\text{tw}(F_k(K_n)) = \Theta(n^k)$ . In the remainder of this section, however, our aim is to establish a considerably sharper upper bound on  $\text{tw}(F_k(K_n))$ , as presented in [Theorem 4](#). Notably, this upper bound is exact for  $k = 2$ , and, as shown by computational experimentation, for certain small values of  $n$  when  $k = 3$ .

As previously mentioned, we assume  $2 \leq k \leq \frac{n}{2}$ , since  $F_1(K_n) \cong K_n$  and  $F_k(K_n) \cong F_{n-k}(K_n)$ . To obtain the upper bound on  $\text{tw}(F_k(K_n))$ , we generalize the path decomposition of  $F_2(K_n)$  given in [Section 3](#).

Before proceeding, we give some notation used throughout this section. When considering an  $r$ -subset  $\{a_1, \dots, a_r\}$  with  $r \geq 2$ , we assume that  $a_1 < \dots < a_r$ . Given two  $r$ -subsets  $X = \{x_1, \dots, x_r\}$  and  $Y = \{y_1, \dots, y_r\}$ , we write  $X \leq Y$  to mean that  $(x_1, \dots, x_r) \leq (y_1, \dots, y_r)$  in the *lexicographic order*; that is, there exists an index  $i \in [r]$  such that  $x_j = y_j$  for all  $j < i$  and  $x_i < y_i$ . For every  $A := \{a_1, \dots, a_k\} \in \binom{[n]}{k}$ , let  $A_s = \{a_1, \dots, a_{k-1}\}$  and  $A_t = \{a_2, \dots, a_k\}$ ; thus,  $A_s, A_t \in \binom{[n]}{k-1}$ .

We now define a path decomposition  $(P, \mathcal{V})$  of  $F_k(K_n)$  as follows:

- Let  $P$  be the path with  $V(P) := \{X \in \binom{[n-1]}{k-1}\}$ , where two subsets  $X$  and  $Y$  are adjacent if they are consecutive in the lexicographic order of  $\binom{[n-1]}{k-1}$ .
- For every  $X \in \binom{[n]}{k-1}$ , let

$$V_X := \left\{ A \in \binom{[n]}{k} : A_s \leq X \leq A_t \right\}.$$

Define  $\mathcal{V} := (V_X)_{X \in V(P)}$ .

**Theorem 16.**  $(P, \mathcal{V})$  is a path decomposition of  $F_k(K_n)$ .

**Proof.** Let  $X \in \binom{[n]}{k-1}$  with  $n \in X$ , and let  $Y$  be its predecessor. Let  $A \in V_X$ . Since  $Y \leq X$  and  $A_s \leq X \leq A_t$ , we have that  $Y \leq A_t$ . Given that  $n \in X$  and  $n \notin A_s$ , we have that  $A_s < X$ . The fact that  $Y$  is the predecessor of  $X$  implies that  $A_s \leq Y$ ; thus,  $A \in V_Y$ . Therefore, if  $X_1, \dots, X_m = X$  is a sequence of consecutive elements in  $\binom{[n]}{k-1}$  in lexicographic order such that  $n \notin X_1$  and  $n \in X_i$  for all  $2 \leq i \leq m$ , then

$$V_X = V_{X_m} \subseteq V_{X_{m-1}} \subseteq \dots \subseteq V_{X_1}. \quad (5)$$

For every  $A \in \binom{[n]}{k}$ , we have that  $A \in V_{A_s}$  and  $A_s \in V(P)$ . Thus,

$$V(F_k(G)) = \bigcup_{X \in \binom{[n-1]}{k-1}} V_X = \bigcup_{X \in V(P)} V_X.$$

Let  $AB$  be an edge of  $F_k(G)$ . Note that  $A \cap B \in \binom{[n]}{k-1}$  and  $A, B \in V_{A \cap B}$ . Thus, by [\(5\)](#),

$$A, B \in V_X \text{ for some vertex } X \in V(P).$$

Let  $A \in V_X \cap V_Y$ , assuming without loss of generality that  $X \leq Y$ , and let  $Z$  be a vertex on the path from  $X$  to  $Y$  in  $P$ , so  $X \leq Z \leq Y$ . By definition,  $A_s \leq X$  and  $Y \leq A_t$ . Since  $A_s \leq X \leq Z \leq Y \leq A_t$ , it follows that  $A \in V_Z$ . This shows that

$$V_X \cap V_Y \subseteq V_Z.$$

Therefore,  $(P, \mathcal{V})$  is a path decomposition of  $F_k(K_n)$ .  $\square$

We now give the exact cardinality of a bag  $V_X \in \mathcal{V}$ .

**Proposition 17.** If  $X = \{x_1, \dots, x_{k-1}\} \in V(P)$ , then

$$|V_X| = x_1 \sum_{i=1}^k \binom{n-x_i}{k-i} - \sum_{i=1}^{k-1} \binom{n-x_{i+1}+1}{k-i},$$

where for convenience we take  $x_k = n$ .

**Proof.** Let  $A := \{a_1, a_2, \dots, a_k\} \in V(F_k(K_n))$ . Our objective is to determine the number of possibilities for the  $a_i$ 's so that  $A \in V_X$ . Note that  $a_1 \leq x_1$  as otherwise  $A_s > X$ , and  $A$  would not be in  $V_X$ . Therefore,  $a_1 < x_1$  or  $a_1 = x_1$ .

- Suppose that  $a_1 < x_1$ ; thus,  $A_s \leq X$  in this case. It remains to establish the conditions so that  $X \leq A_t$ . Note that  $X \leq A_t$  if and only if either  $a_i = x_{i-1}$  for all  $2 \leq i \leq k$ , or for some  $2 \leq i \leq k$ , we have that

$$a_j = x_{j-1} \text{ for all } 2 \leq j < i, \text{ and } a_i > x_{i-1}.$$

In the latter case there are  $\binom{n-x_{i-1}}{k-(i-1)}$  choices for the remaining  $\{a_i, \dots, a_k\}$ . We take  $\binom{n-x_k}{0} = 1$  to represent the possibility where  $a_i = x_{i-1}$  for all  $i = 2, \dots, k$ ; considering that there are  $x_1 - 1$  choices for  $a_1$  so that  $a_1 < x_1$ , we have that there in total

$$(x_1 - 1) \left[ \binom{n-x_1}{k-1} + \binom{n-x_2}{k-2} + \dots + \binom{n-x_k}{0} \right] = (x_1 - 1) \sum_{i=1}^k \binom{n-x_i}{k-i} \quad (6)$$

possible choices for  $\{a_1, \dots, a_k\}$  so that  $a_1 < x_1$  and  $A \in V_X$ .

- Suppose that  $a_1 = x_1$ ; since  $a_2 > a_1 = x_1$ , we have that  $X \leq A_t$  in this case. It remains to establish the conditions so that  $A_s \leq X$ . Note that  $A_s \leq X$  if and only if either  $a_i = x_i$  for all  $2 \leq i \leq k-1$ , or for some  $2 \leq i \leq k-1$ , we have that

$$a_j = x_j \text{ for all } 2 \leq j < i, \text{ and } x_{i-1} < a_i < x_i.$$

In the latter case, there are  $\binom{n-a_i}{k-i}$  choices for the remaining  $\{a_{i+1}, \dots, a_k\}$ . Recall that for convenience we take  $x_k = n$ . Thus, we have in total

$$\sum_{x_1 < a_2 < x_2} \binom{n-a_2}{k-2} + \sum_{x_2 < a_3 < x_3} \binom{n-a_3}{k-3} + \dots + \sum_{x_{k-1} < a_k < x_k} \binom{n-a_k}{0} + 1 \quad (7)$$

possible choices for  $\{a_1, \dots, a_k\}$  so that  $a_1 = x_1$  and  $A \in V_X$ . Observe that for  $1 < i \leq k$ , we have that

$$\begin{aligned} \sum_{x_{i-1} < a_i < x_i} \binom{n-a_i}{k-i} &= \sum_{j=n-x_i+1}^{n-x_{i-1}-1} \binom{j}{k-i} \\ &= \sum_{j=1}^{n-x_{i-1}-1} \binom{j}{k-i} - \sum_{j=1}^{n-x_i} \binom{j}{k-i} \\ &= \binom{n-x_{i-1}}{k-i+1} - \binom{n-x_i}{k-i+1}, \end{aligned}$$

where the last equality follows from the hockey-stick identity.<sup>1</sup> In particular, if  $x_i = x_{i-1} + 1$ , we have  $\sum_{x_{i-1} < a_i < x_i} \binom{n-a_i}{k-i} = 0$ .

Rephrasing Eq. (7), we obtain

$$\begin{aligned} 1 + \sum_{i=2}^k \sum_{x_{i-1} < a_i < x_i} \binom{n-a_i}{k-i} &= 1 + \sum_{i=2}^k \left[ \binom{n-x_{i-1}}{k-i+1} - \binom{n-x_i}{k-i+1} \right] \\ &= 1 + \sum_{i=1}^{k-1} \left[ \binom{n-x_i}{k-i} - \binom{n-x_{i+1}}{k-i} \right] \\ &= \sum_{i=1}^k \binom{n-x_i}{k-i} - \sum_{i=1}^{k-1} \binom{n-x_{i+1}}{k-i} \end{aligned} \quad (8)$$

Combining Eqs. (6) and (8), we have that

$$|V_X| = x_1 \sum_{i=1}^k \binom{n-x_i}{k-i} - \sum_{i=1}^{k-1} \binom{n-x_{i+1}}{k-i}.$$

This concludes the proof.  $\square$

To determine the bag  $V_X$  of maximum size, we partition  $\mathcal{V}$  into families, and determine the bag of maximum size for each of these families.

For each  $i \in \{1, \dots, n-k+1\}$ , let

$$\mathcal{V}_i := \{V_X \in \mathcal{V} : X = \{x_1, \dots, x_{k-1}\} \text{ and } x_1 = i\}.$$

We now define a family  $\mathcal{M}_i$  of  $(k-1)$ -subsets of  $[n-1]$ , which will serve as a subset of indices for bags in  $\mathcal{V}_i$ . Specifically, let  $\mathcal{M}_i$  consist of all sets  $X = \{x_1, \dots, x_{k-1}\} \in \binom{[n-1]}{k-1}$  such that  $x_1 = i$  and, for each  $j = 2, \dots, k-1$ , the following conditions hold:

- (a)  $x_j = i + j - 1$  whenever  $n - (k-j)i + 1 < i + j - 1$ , and,
- (b)  $x_j \in \{n - (k-j)i, n - (k-j)i + 1\}$ , otherwise.

If for some  $2 \leq j \leq k-2$  we have that  $n - (k-j)i + 1 \geq i + j - 1$ , then  $n - (k-(j+1))i + 1 \geq i + (j+1) - 1$ . This implies that the sequence  $x_1, \dots, x_{k-1}$  is constructed by first applying rule (a) and afterwards (possibly) only applying rule (b). In particular, the sequence is increasing and thus defines a valid element of  $\binom{[n-1]}{k-1}$ .

Therefore, each  $X \in \mathcal{M}_i$  indexes a bag  $V_X \in \mathcal{V}_i$ , and we have

$$\{V_X : X \in \mathcal{M}_i\} \subseteq \mathcal{V}_i.$$

<sup>1</sup>  $\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$  for all natural numbers  $n \geq r$ .

**Lemma 18.** Let  $i \in \{1, \dots, n - k + 1\}$  and  $X \in \mathcal{V}_i$ . Then

$$|V_X| = \max_{Y \in \mathcal{V}_i} |V_Y| \text{ if and only if } X \in \mathcal{M}_i.$$

**Proof.** Let  $X \in \mathcal{V}_i$  and let  $Y = \{y_1, \dots, y_{k-1}\} \in \mathcal{M}_i$ . We define the distance between  $X$  and  $Y$  as

$$d(X, Y) := \sum_{j=1}^{k-1} |x_j - y_j|.$$

We iteratively construct a sequence

$$X = X_0, X_1, \dots, X_m = Y$$

of elements in  $\mathcal{V}_i$ , such that:

- (1)  $|V_{X_0}| \leq |V_{X_1}| \leq \dots \leq |V_{X_m}|$ ;
- (2)  $d(X_l, Y) = d(X_{l-1}, Y) - 1$  for all  $2 \leq l \leq m$ ; and
- (3)  $|V_{X_0}| < |V_{X_m}|$  if  $X \notin \mathcal{M}_i$ .

Suppose that  $X_{l-1} = \{x_1, \dots, x_{k-1}\}$  for some  $1 \leq l \leq m - 1$  is the last element so constructed and that  $X_{l-1} \neq Y$ . If  $x_j < y_j$  for some  $2 \leq j \leq m$ , then let  $j^*$  be the largest index such that  $x_{j^*} < y_{j^*}$ . Otherwise, let  $j^*$  be the smallest index such that  $x_{j^*} > y_{j^*}$ . Observe that, in both cases,  $j^* \neq 1$ .

Let

$$X_l := \begin{cases} X_{l-1} \setminus \{x_{j^*}\} \cup \{x_{j^*} + 1\}, & \text{if } x_{j^*} < y_{j^*}, \\ X_{l-1} \setminus \{x_{j^*}\} \cup \{x_{j^*} - 1\}, & \text{if } x_{j^*} > y_{j^*}. \end{cases}$$

Suppose  $x_{j^*} < y_{j^*}$ . Since  $x_{j^*} < y_{j^*} \leq n - 1$ , we have  $x_{j^*} + 1 \leq n - 1$ . Furthermore, by the choice of  $j^*$ , we have  $x_{j^*+1} = y_{j^*+1} \geq y_{j^*} + 1 > x_{j^*} + 1$ . Therefore,  $x_{j^*} + 1 \notin X_{l-1}$ , ensuring that  $X_l$  is well-defined. Now, suppose that  $x_{j^*} > y_{j^*}$ . By the choice of  $j^*$ , we know that  $x_{j^*-1} = y_{j^*-1} \leq y_{j^*} - 1 < x_{j^*} - 1$ . Hence,  $x_{j^*} - 1 \notin X_{l-1}$ , implying that  $X_l$  is well-defined. In either case,  $X_l \in \mathcal{V}_i$  and

$$d(X_l, Y) = d(X_{l-1}, Y) - 1.$$

This proves (2). We now show (1).

Suppose that  $x_{j^*} < y_{j^*}$ . By [Proposition 17](#), we have that

$$\begin{aligned} |V_{X_l}| - |V_{X_{l-1}}| &= \left[ x_1 \sum_{\substack{j=1 \\ j \neq j^*}}^k \binom{n - x_j}{k - j} - \sum_{\substack{j=1 \\ j \neq j^* - 1}}^{k-1} \binom{n - x_{j+1} + 1}{k - j} + x_1 \binom{n - (x_{j^*} + 1)}{k - (j^* - 1)} - \binom{n - (x_{j^*} + 1) + 1}{k - (j^* - 1)} \right] \\ &\quad - \left[ x_1 \sum_{\substack{j=1 \\ j \neq j^*}}^k \binom{n - x_j}{k - j} - \sum_{\substack{j=1 \\ j \neq j^* - 1}}^{k-1} \binom{n - x_{j+1} + 1}{k - j} + x_1 \binom{n - x_{j^*}}{k - (j^* - 1)} - \binom{n - x_{j^*} + 1}{k - (j^* - 1)} \right] \\ &= x_1 \left[ \binom{n - x_{j^*} - 1}{k - j^*} - \binom{n - x_{j^*}}{k - j^*} \right] - \left[ \binom{n - x_{j^*}}{k - j^* + 1} - \binom{n - x_{j^*} + 1}{k - j^* + 1} \right] \\ &= \binom{n - x_{j^*}}{k - j^*} - x_1 \binom{n - x_{j^*} - 1}{k - j^* - 1} \\ &= \binom{n - x_{j^*} - 1}{k - j^* - 1} \left[ \frac{(n - (k - j^*)x_1) - x_{j^*}}{k - j^*} \right] \\ &\geq 0. \end{aligned}$$

Where in the last line

$$\text{we have equality if and only if } x_{j^*} = n - (k - j^*)x_1. \quad (9)$$

Suppose that  $x_{j^*} > y_{j^*}$ . By [Proposition 17](#), we have that

$$|V_{X_l}| - |V_{X_{l-1}}| = \left[ x_1 \sum_{\substack{j=1 \\ j \neq j^*}}^k \binom{n - x_j}{k - j} - \sum_{\substack{j=1 \\ j \neq j^* - 1}}^{k-1} \binom{n - x_{j+1} + 1}{k - j} + x_1 \binom{n - (x_{j^*} - 1)}{k - (j^* - 1)} - \binom{n - (x_{j^*} - 1) + 1}{k - (j^* - 1)} \right]$$

$$\begin{aligned}
 & - \left[ x_1 \sum_{\substack{j=1 \\ j \neq j^*}}^k \binom{n-x_j}{k-j} - \sum_{\substack{j=1 \\ j \neq j^*-1}}^{k-1} \binom{n-x_{j+1}+1}{k-j} + x_1 \binom{n-x_{j^*}}{k-(j^*-1)} - \binom{n-x_{j^*}+1}{k-(j^*-1)} \right] \\
 & = x_1 \left[ \binom{n-x_{j^*}+1}{k-j^*} - \binom{n-x_{j^*}}{k-j^*} \right] - \left[ \binom{n-x_{j^*}+2}{k-j^*+1} - \binom{n-x_{j^*}+1}{k-j^*+1} \right] \\
 & = x_1 \binom{n-x_{j^*}}{k-j^*-1} - \binom{n-x_{j^*}+1}{k-j^*} \\
 & = x_1 \binom{n-x_{j^*}}{k-j^*-1} - \frac{n-x_{j^*}+1}{k-j^*} \binom{n-x_{j^*}}{k-j^*-1} \\
 & = \binom{n-x_{j^*}}{k-j^*-1} \left[ x_1 - \frac{n-x_{j^*}+1}{k-j^*} \right] \\
 & = \binom{n-x_{j^*}}{k-j^*-1} \left[ \frac{x_{j^*} - (n-(k-j^*)x_1+1)}{k-j^*} \right] \\
 & \geq 0,
 \end{aligned}$$

Where in the last line

$$\text{we have equality if and only if } x_{j^*} = n - (k-j^*)x_1 + 1. \quad (10)$$

This proves (1).

Suppose that  $X \notin \mathcal{M}_i$ . To apply the previous comparison between  $|V_{X_{l-1}}|$  and  $|V_{X_l}|$ , we set  $X = \{x_1, \dots, x_{k-1}\} = X_{l-1}$  and take  $l = 1$ . We will show that  $|V_{X_{l-1}}| < |V_{X_l}|$ , and thus, by (1), obtain  $|V_X| < |V_Y|$ , as desired. Let  $2 \leq j^* \leq k$  be the smallest index such that  $x_{j^*}$  does not satisfy rules (a) and (b).

Suppose that  $x_{j^*} < y_{j^*}$ . If  $y_{j^*}$  was constructed by applying rule (a), then

$$y_{j^*} = y_1 + j^* - 1 = x_1 + j^* - 1 \leq x_{j^*},$$

a contradiction. Thus,  $y_{j^*}$  was constructed by rule (b), and hence  $y_{j^*} \in \{n - (k-j^*)x_1, n - (k-j^*)x_1 + 1\}$ . Since  $x_{j^*} < y_{j^*}$  and  $x_{j^*}$  does not satisfy rules (a) or (b), it follows that

$$x_{j^*} < n - (k-j^*)x_1.$$

By (9), we have that

$$|V_{X_l}| > |V_{X_{l-1}}|.$$

Suppose that  $x_{j^*} > y_{j^*}$ . Suppose that  $y_{j^*}$  was produced by rule (a). Thus,

$$n - (k-j^*)i + 1 < i + j^* - 1 = y_{j^*} < x_{j^*}.$$

By (10), we have that

$$|V_{X_l}| > |V_{X_{l-1}}|.$$

Now suppose that  $y_{j^*}$  was produced by rule (b), so  $y_{j^*} \in \{n - (k-j^*)x_1, n - (k-j^*)x_1 + 1\}$ . Since  $x_{j^*} > y_{j^*}$  and  $x_{j^*}$  does not satisfy rules (a) or (b), it follows that

$$x_{j^*} > n - (k-j^*)i + 1.$$

Again, by (10), we conclude that

$$|V_{X_l}| > |V_{X_{l-1}}|.$$

This proves (3).  $\square$

**Lemma 19.** Let  $X := \{x_1, \dots, x_{k-1}\} \in V(P)$  that maximizes  $|V_X|$ . Then  $x_1$  is equal to

$$\left\lfloor \frac{n+1}{k} \right\rfloor \text{ or } \left\lceil \frac{n}{k} \right\rceil.$$

**Proof.** Suppose for a contradiction that  $x_1 \neq \left\lfloor \frac{n+1}{k} \right\rfloor, \left\lceil \frac{n}{k} \right\rceil$ . We show that there exists  $Y \in V(P)$  such that  $|V_Y| > |V_X|$ . If  $k$  divides  $n$  or  $n+1$ , then  $\left\lfloor \frac{n+1}{k} \right\rfloor = \left\lceil \frac{n}{k} \right\rceil$ ; and if  $k$  divides neither, then  $\left\lfloor \frac{n+1}{k} \right\rfloor = \left\lfloor \frac{n}{k} \right\rfloor < \left\lceil \frac{n}{k} \right\rceil$ ; therefore,  $\left\lfloor \frac{n+1}{k} \right\rfloor \leq \left\lceil \frac{n}{k} \right\rceil$ .

- Suppose that  $x_1 < \left\lfloor \frac{n+1}{k} \right\rfloor$ . We now show that  $x_2 > x_1 + 1$ . Set  $i = x_1$  and  $j = 2$ . By Lemma 18, we have  $X \in \mathcal{M}_i$ , so  $x_2$  is constructed according to either rule (a) or rule (b). Suppose first that  $n - (k-j)i \leq i + j - 1$ , so  $i \geq \frac{n-1}{k-1}$ . On the

other hand, since  $i = x_1 < \lfloor \frac{n+1}{k} \rfloor \leq \frac{n+1}{k}$ , it follows that

$$\frac{n-1}{k-1} \leq i < \frac{n+1}{k},$$

which implies  $k > \frac{n+1}{2}$ , a contradiction to the assumption  $k \leq \frac{n}{2}$ . Therefore, we must have  $n - (k-j)i > i + j - 1$ , which means that  $x_2$  was constructed by rule (b). Thus,

$$x_2 \geq n - (k-j)i > i + j - 1 = x_1 + 1,$$

as desired.

Let

$$Y := \{y_1, \dots, y_{k-1}\} = (X \setminus \{x_1\}) \cup \{x_1 + 1\};$$

thus,  $y_1 = x_1 + 1$  and  $y_j = x_j$  for every  $j \in \{2, \dots, k-1\}$ . Note that this is well-defined because we have already established that  $x_2 > x_1 + 1$ , ensuring that the choice of  $y_1 = x_1 + 1$  is valid. By Proposition 17, we have that

$$\begin{aligned} |V_Y| - |V_X| &= y_1 \sum_{i=1}^k \binom{n-y_i}{k-i} - \sum_{i=1}^{k-1} \binom{n-y_{i+1}+1}{k-i} \\ &\quad - x_1 \sum_{i=1}^k \binom{n-x_i}{k-i} + \sum_{i=1}^{k-1} \binom{n-x_{i+1}+1}{k-i} \\ &= y_1 \sum_{i=1}^k \binom{n-y_i}{k-i} - x_1 \sum_{i=1}^k \binom{n-x_i}{k-i} \\ &= (x_1 + 1) \binom{n-x_1-1}{k-1} + (x_1 + 1) \sum_{i=2}^k \binom{n-x_i}{k-i} \\ &\quad - x_1 \binom{n-x_1}{k-1} - x_1 \sum_{i=2}^k \binom{n-x_i}{k-i} \\ &= (x_1 + 1) \binom{n-x_1-1}{k-1} + \sum_{i=2}^k \binom{n-x_i}{k-i} - x_1 \binom{n-x_1}{k-1} \\ &> (x_1 + 1) \binom{n-x_1-1}{k-1} - x_1 \binom{n-x_1}{k-1} \\ &= \frac{(x_1 + 1)(n-x_1-1)!}{(k-1)!(n-x_1-k)!} - \frac{x_1(n-x_1)!}{(k-1)!(n-x_1-k+1)!} \\ &= \frac{(n-x_1-1)!}{(k-1)!(n-x_1-k)!} \left[ (x_1 + 1) - \frac{x_1(n-x_1)}{n-x_1-k+1} \right] \\ &= \binom{n-x_1-1}{k-1} \left[ \frac{n+1-k(x_1+1)}{n-x_1-k+1} \right] \\ &\geq 0. \end{aligned}$$

Here, the strict inequality comes from the fact that

$$\sum_{i=2}^k \binom{n-x_i}{k-i} \geq \binom{n-x_k}{k-k} = \binom{0}{0} = 1,$$

since  $x_k = n$ . On the other hand, the last inequality is obtained from the following. Since  $x_1 + 1 \leq \lfloor \frac{n+1}{k} \rfloor$ , we have that

$$n+1-k(x_1+1) \geq n+1-k \left\lfloor \frac{n+1}{k} \right\rfloor \geq n+1-(n+1)=0.$$

Moreover, recall that  $n \notin X$ . This implies that  $x_1 \leq x_{k-1} - (k-2) \leq n-k+1$ , and hence  $n-x_1-k+1 \geq 0$ . We conclude that  $|V_Y| > |V_X|$ .

- Suppose now that  $x_1 > \lceil \frac{n}{k} \rceil$ . We choose

$$Y := \{y_1, \dots, y_{k-1}\} = (X \setminus \{x_1\}) \cup \{x_1 - 1\};$$

thus  $y_1 = x_1 - 1 \geq \lceil \frac{n}{k} \rceil$ , and  $y_i = x_i$  for  $i = 2, \dots, k-1$ . Since  $x_i \geq x_1 + i - 1$ , for all  $i = 2, \dots, k$ , we have that

$$\begin{aligned} \binom{n-x_1}{k-2} &= \binom{n-x_1-1}{k-2} + \binom{n-x_1-1}{k-3} \\ &= \binom{n-x_1-1}{k-2} + \binom{n-x_1-2}{k-3} + \binom{n-x_1-2}{k-4} \\ &\quad \vdots \\ &= \binom{n-x_1-1}{k-2} + \binom{n-x_1-2}{k-3} + \dots + \binom{n-x_1-k+1}{0} \\ &= \sum_{i=2}^k \binom{n-(x_1+i-1)}{k-i} \\ &\geq \sum_{i=2}^k \binom{n-x_i}{k-i}. \end{aligned}$$

Furthermore, if  $x_1 - 1 = \lceil \frac{n}{k} \rceil$ , we have that  $x_2 > x_1 + 1$ , and so we obtain that

$$\binom{n-x_1}{k-2} > \sum_{i=2}^k \binom{n-x_i}{k-i}. \quad (11)$$

We now compute  $|V_X| - |V_Y|$ . By Proposition 17, we have that

$$\begin{aligned} |V_X| - |V_Y| &= x_1 \sum_{i=1}^k \binom{n-x_i}{k-i} - \sum_{i=1}^{k-1} \binom{n-x_{i+1}+1}{k-i} \\ &\quad - y_1 \sum_{i=1}^k \binom{n-y_i}{k-i} + \sum_{i=1}^{k-1} \binom{n-y_{i+1}+1}{k-i} \\ &= x_1 \sum_{i=1}^k \binom{n-x_i}{k-i} - y_1 \sum_{i=1}^k \binom{n-y_i}{k-i} \\ &= x_1 \binom{n-x_1}{k-1} + x_1 \sum_{i=2}^k \binom{n-x_i}{k-i} \\ &\quad - (x_1-1) \binom{n-x_1+1}{k-1} - (x_1-1) \sum_{i=2}^k \binom{n-x_i}{k-i} \\ &= x_1 \binom{n-x_1}{k-1} + \sum_{i=2}^k \binom{n-x_i}{k-i} - (x_1-1) \binom{n-x_1+1}{k-1} \\ &= x_1 \binom{n-x_1}{k-1} - (x_1-1) \left( \binom{n-x_1}{k-1} + \binom{n-x_1}{k-2} \right) + \sum_{i=2}^k \binom{n-x_i}{k-i} \\ &= \binom{n-x_1}{k-1} - (x_1-1) \binom{n-x_1}{k-2} + \sum_{i=2}^k \binom{n-x_i}{k-i} \\ &\leq \binom{n-x_1}{k-1} + \binom{n-x_1}{k-2} - (x_1-1) \binom{n-x_1}{k-2} \\ &= \binom{n-x_1+1}{k-1} - (x_1-1) \binom{n-x_1}{k-2} \\ &= \frac{n-x_1+1}{k-1} \binom{n-x_1}{k-2} - (x_1-1) \binom{n-x_1}{k-2} \\ &= \binom{n-x_1}{k-2} \left[ \frac{n-x_1+1}{k-1} - (x_1-1) \right] \\ &= \binom{n-x_1}{k-2} \left[ \frac{n-x_1+1-(x_1-1)(k-1)}{k-1} \right] \end{aligned}$$

$$= \binom{n-x_1}{k-2} \left[ \frac{n-k(x_1-1)}{k-1} \right].$$

Thus, if  $x_1 - 1 > \lceil \frac{n}{k} \rceil$ , it follows that  $|V_X| < |V_Y|$ . On the other hand, if  $x_1 - 1 = \lceil \frac{n}{k} \rceil$ , then we have  $n - k(x_1 - 1) \leq 0$ . Furthermore, by Eq. (11), the inequality ( $\leq$ ) in our sequence of steps becomes a strict inequality ( $<$ ), ensuring that  $|V_X| < |V_Y|$ . In either case, we conclude that  $|V_X| < |V_Y|$ , as claimed. This completes the proof.  $\square$

We are ready to prove our upper bound on  $\text{pw}(F_k(K_n))$  as stated in Theorem 4.

**Theorem 4.** Let  $2 \leq k \leq n/2$ . Then  $\text{pw}(F_k(K_n))$  is at most

$$\max \left\{ \left\lfloor \frac{n+1}{k} \right\rfloor \binom{n - \lfloor \frac{n+1}{k} \rfloor}{k-1} + \left\lfloor \frac{n+1}{k} \right\rfloor \sum_{i=2}^k \binom{(k-i)\lfloor \frac{n+1}{k} \rfloor}{k-i} - \sum_{i=1}^{k-1} \binom{(k-i-1)\lfloor \frac{n+1}{k} \rfloor + 1}{k-i}, \right. \\ \left. \left\lceil \frac{n}{k} \right\rceil \binom{n - \lceil \frac{n}{k} \rceil}{k-1} + \left\lceil \frac{n}{k} \right\rceil \sum_{i=2}^k \binom{(k-i)\lceil \frac{n}{k} \rceil}{k-i} - \sum_{i=1}^{k-1} \binom{(k-i-1)\lceil \frac{n}{k} \rceil + 1}{k-i} \right\} - 1.$$

Further, if  $k$  divides  $n$  or  $n+1$ , these two values are equal.

**Proof.** Recall that the width of  $(P, \nu)$  is an upper bound for  $\text{tw}(F_k(K_n))$ . Let  $X := \{x_1, \dots, x_{k-1}\}$  and  $Y := \{y_1, \dots, y_{k-1}\}$ , where

$$x_i := \begin{cases} \lfloor \frac{n+1}{k} \rfloor & \text{if } i = 1, \\ n - (k-i)x_1 & \text{otherwise,} \end{cases} \quad \text{and} \quad y_i := \begin{cases} \lceil \frac{n}{k} \rceil & \text{if } i = 1, \\ n - (k-i)y_1 & \text{otherwise.} \end{cases}$$

By Lemmas 18 and 19 it follows that  $\text{width}(P, \nu) = \max\{|V_X|, |V_Y|\} - 1$ . Then, by Proposition 17, we have that  $\text{width}(P, \nu)$  is equal to

$$\max \left\{ \left\lfloor \frac{n+1}{k} \right\rfloor \binom{n - \lfloor \frac{n+1}{k} \rfloor}{k-1} + \left\lfloor \frac{n+1}{k} \right\rfloor \sum_{i=2}^k \binom{(k-i)\lfloor \frac{n+1}{k} \rfloor}{k-i} - \sum_{i=1}^{k-1} \binom{(k-i-1)\lfloor \frac{n+1}{k} \rfloor + 1}{k-i}, \right. \\ \left. \left\lceil \frac{n}{k} \right\rceil \binom{n - \lceil \frac{n}{k} \rceil}{k-1} + \left\lceil \frac{n}{k} \right\rceil \sum_{i=2}^k \binom{(k-i)\lceil \frac{n}{k} \rceil}{k-i} - \sum_{i=1}^{k-1} \binom{(k-i-1)\lceil \frac{n}{k} \rceil + 1}{k-i} \right\} - 1.$$

Finally, note that if  $k$  divides  $n$  or  $n+1$ , then  $\lfloor \frac{n+1}{k} \rfloor = \lceil \frac{n}{k} \rceil$ .  $\square$

We conclude this section by establishing the precise value of our upper bound on  $\text{pw}(F_3(K_n))$ .

**Corollary 5.** Let  $n \geq 6$ . Then

$$\text{pw}(F_3(K_n)) \leq \left\lfloor \frac{n}{3} \right\rfloor \binom{n - \lceil \frac{n}{3} \rceil}{2} + \frac{1}{2} \left\lceil \frac{n}{3} \right\rceil \left( \left\lceil \frac{n}{3} \right\rceil + 1 \right) - 2.$$

**Proof.** Let  $X$  and  $Y$  be as defined in the proof of Theorem 4. Notice that  $\lfloor \frac{n+1}{3} \rfloor \leq \lceil \frac{n}{3} \rceil$ , and if equality holds, our upper bound is the desired value. Suppose now that  $\lfloor \frac{n+1}{3} \rfloor < \lceil \frac{n}{3} \rceil$ , implying  $n \equiv 1 \pmod{3}$ . Let  $t := \lfloor \frac{n+1}{3} \rfloor$ , so  $t+1 = \lceil \frac{n}{3} \rceil$  and  $n = 3t+1$ . Then,

$$\begin{aligned} |V_Y| - |V_X| &= (t+1) \left[ \binom{n-t-1}{2} + \binom{t+1}{1} + \binom{0}{0} \right] - \left[ \binom{t+2}{2} + \binom{1}{1} \right] \\ &\quad - t \left[ \binom{n-t}{2} + \binom{t}{1} + \binom{0}{0} \right] + \left[ \binom{t+1}{2} + \binom{1}{1} \right] \\ &= (t+1) \left[ \binom{n-t-1}{2} + t+2 \right] - \binom{t+2}{2} - t \left[ \binom{n-t}{2} + t+1 \right] + \binom{t+1}{2} \\ &= (t+1) \binom{n-t-1}{2} + (t+1)(t+2) - \binom{t+2}{2} - t \binom{n-t}{2} - t(t+1) + \binom{t+1}{2} \\ &= (t+1) \binom{n-t-1}{2} - t \binom{n-t}{2} + t+1 \\ &= \binom{n-t-1}{2} - t(n-t-2) + 1 \\ &= (n-t-2) \left( \frac{n-t-1}{2} - t \right) + 1 \end{aligned}$$



$$\begin{aligned} &= (n - t - 2) \frac{n - 3t - 1}{2} + 1 \\ &= 1. \end{aligned}$$

Hence,  $|V_Y| > |V_X|$ . Finally, we have

$$\begin{aligned} \text{pw}(F_3(K_n)) &\leq |V_Y| - 1 = \left\lceil \frac{n}{3} \right\rceil \left[ \binom{n - \left\lceil \frac{n}{3} \right\rceil}{2} + \left\lceil \frac{n}{3} \right\rceil + 1 \right] - \left[ \left( \left\lceil \frac{n}{3} \right\rceil + 1 \right) + 1 \right] - 1 \\ &= \left\lceil \frac{n}{3} \right\rceil \left( \binom{n - \left\lceil \frac{n}{3} \right\rceil}{2} + \frac{1}{2} \left\lceil \frac{n}{3} \right\rceil \left( \left\lceil \frac{n}{3} \right\rceil + 1 \right) - 2 \right). \quad \square \end{aligned}$$

## 5. Summary of results and conjectures

In this paper, we have established tight asymptotic lower and upper bounds on the treewidth of token graphs for three families of graphs. The first two of them are special cases of trees (stars and paths), for which we obtained that the treewidth of their  $k$ -token graphs, for a fixed integer  $k$ , is  $\Theta(n^{k-1})$ . Additionally, we have shown that the treewidth of the  $k$ -token graph of the complete graph  $K_n$  (Johnson graph) is, for a fixed integer  $k$ , equal to  $\Theta(n^k)$ . These results lead us to formulate the following conjectures:

**Conjecture 20.** *If  $2 \leq k \leq n/2$  is a fixed integer and  $G$  is a tree on  $n$  vertices, then*

$$\text{tw}(F_k(G)) \in \Theta(n^{k-1}).$$

**Conjecture 21.** *If  $2 \leq k \leq n/2$  is a fixed integer, then the tree decomposition of the Johnson graph given in Section 4 is optimal, and therefore*

$$\text{pw}(F_k(K_n)) = \text{tw}(F_k(K_n)).$$

Furthermore, the treewidth of the Johnson graph is equal to the upper bound given in Theorem 4.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

No data was used for the research described in the article.

## References

- [1] Y. Alavi, M. Behzad, P. Erdős, D.R. Lick, Double vertex graphs, *J. Comb. Inf. Syst. Sci.* 16 (1) (1991) 37–50.
- [2] O. Amini, J. Kool, A spectral lower bound for the divisorial gonality of metric graphs, *Int. Math. Res. Not. IMRN* 2016 (8) (2016) 2423–2450.
- [3] K. Audenaert, C. Godsil, G. Royle, T. Rudolph, Symmetric squares of graphs, *J. Combin. Theory Ser. B* 97 (1) (2007) 74–90, <http://dx.doi.org/10.1016/j.jctb.2006.04.002>.
- [4] S. Barik, P. Verma, Spectral properties of token graphs, *Linear Algebra Appl.* 687 (2024) 181–206.
- [5] H. Bodlaender, A tourist guide through treewidth, *Acta Cybernet.* 11 (1–2) (1993).
- [6] J. Bondy, U. Murty, *Graph Theory with Applications*, Macmillan Press, London, 1976.
- [7] P. Caputo, T. Liggett, T. Richthammer, Proof of Aldous’ spectral gap conjecture, *J. Amer. Math. Soc.* 23 (3) (2010) 831–851.

- [8] L.S. Chandran, C. Subramanian, A spectral lower bound for the treewidth of a graph and its consequences, *Inform. Process. Lett.* 87 (4) (2003) 195–200.
- [9] C. Dalfó, F. Duque, R. Fabila-Monroy, M. Fiol, C. Huemer, A. Trujillo-Negrete, F. Zaragoza Martínez, On the Laplacian spectra of token graphs, *Linear Algebra Appl.* 625 (2021) 322–348, <http://dx.doi.org/10.1016/j.laa.2021.05.005>.
- [10] N.M.M. De Abreu, Old and new results on algebraic connectivity of graphs, *Linear Algebra Appl.* 423 (1) (2007) 53–73.
- [11] R. Fabila-Monroy, D. Flores-Peñaloza, C. Huemer, F. Hurtado, J. Urrutia, D.R. Wood, Token graphs, *Graphs Combin.* 28 (3) (2012) 365–380, <http://dx.doi.org/10.1007/s00373-011-1055-9>.
- [12] R.J. Faudree, R.H. Schelp, Path Ramsey numbers in multicolorings, *J. Combin. Theory Ser. B* 19 (2) (1975) 150–160.
- [13] D.J. Harvey, D.R. Wood, Treewidth of the line graph of a complete graph, *J. Graph Theory* 79 (1) (2015) 48–54.
- [14] R. Hickingbotham, D.R. Wood, Structural properties of graph products, *J. Graph Theory* 109 (2) (2025) 107–136, <http://dx.doi.org/10.1002/jgt.23023>.
- [15] G.L. Johns, Generalized Distance in Graphs (Ph.D. thesis), Western Michigan University, 1988, URL <https://scholarworks.wmich.edu/dissertations/2180>.
- [16] B.O. Lucena, Dynamic Programming, Tree-Width and Computation on Graphical Models (Ph.D. thesis), Brown University, 2002, URL <https://dokumen.tips/documents/dynamic-programming-tree-width-and-computation-on-graphical-models.html?page=1>.
- [17] T. Rudolph, Constructing physically intuitive graph invariants, 2002, ArXiv Preprint ArXiv:Quant-Ph/0206068. URL <https://arxiv.org/abs/quant-ph/0206068>.
- [18] P.D. Seymour, R. Thomas, Graph searching and a min-max theorem for tree-width, *J. Combin. Theory Ser. B* 58 (1) (1993) 22–33.
- [19] B.W. Zhu, J. Liu, D.R. Lick, Y. Alavi,  $n$ -tuple vertex graphs, in: *Proceedings of the Twenty-Third Southeastern International Conference on Combinatorics, Graph Theory, and Computing* (Boca Raton, FL, 1992), vol. 89, 1992, pp. 97–106.